Time-JEPA

TRAINING

PHYSICAL TIME-SERIES PREPARATION MODULE

Physical time series → Time alignment → Tubelet pooling → State normalization → Tubelet-level state values

PHYSICAL TIME-SERIES SUPERVISION MODULE

Full clip → online encoder → Patch tokens → Spatial mean → Auxiliary head → Regressed state values → L_{time}

Pointwise regression

shared

PREDICTIVE VIDEO REPRESENTATION MODULE

Masked clip → online encoder → Predictor → Predicted latent tokens → L_{JEPA}

EMA TARGET PATH

Full clip → EMA target encoder → Latent tokens

EMA update

Joint objective

$$L = L_{JEPA} + \lambda L_{time}$$

VIDEO-ONLY EVALUATION

remove auxiliary head

Video-only input

Frozen video encoder

Task-specific probes

Current-state decoding

Within the clip

Future-state prediction

Multi-horizon 0.2–1.0 s

Kinematic reconstruction

Fixed relations

External physical evaluations

Surprise / contact